

Reproduction and Critical Evaluation of Hidden Markov Models for Speech Recognition with Integrated Pitch and MFCC Features

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Demo Video: [<https://youtu.be/RXa0OmY2A3Q>]

Code Repository: [<https://github.com/PranavGupta26/hmm-speech-emotion>]

1 Introduction

Automatic speech and emotion recognition are central problems in artificial intelligence, with applications spanning human-computer interaction, mental health monitoring, and affective computing. The ability to infer emotional state from speech is particularly valuable in settings where non-verbal communication is the primary signal available to a system.

Hidden Markov Models (HMMs) provide a principled probabilistic framework for sequential data and have remained foundational in automatic speech recognition (ASR) since the 1970s. Mel Frequency Cepstral Coefficients (MFCCs) are the dominant spectral feature in classical ASR pipelines, derived by applying a mel-scale filterbank to the short-time Fourier transform of a speech frame, followed by a discrete cosine transform (DCT) to decorrelate filterbank energies. While MFCCs capture the spectral envelope effectively, they do not encode prosodic information such as fundamental frequency (pitch) or frame-level energy—cues that carry significant discriminative information in emotionally expressive speech.

This paper was selected because it represents a clear, self-contained instance of classical probabilistic sequence modeling applied to a structured emotion recognition task. It offers a well-defined mathematical framework, an implementable feature pipeline, and verifiable claims about model behavior, making it well-suited for a rigorous reproduction study. This report describes the reproduction and critical evaluation of an HMM-based speech emotion recognition system integrating MFCC features with pitch and energy, trained on four emotion classes—happy, sad, angry, and neutral—and evaluated on 40 real-time audio recordings.

2 Paper Overview

2.1 Core Idea and Contributions

The paper proposes augmenting standard MFCC-based HMM speech recognition with prosodic features—specifically pitch and log energy—to improve emotion classification. The central claim is that spectral features alone are insufficient for distinguishing emotionally expressive speech, and that integrating pitch and energy under a scaled composite feature vector improves class-conditional separation. The key contributions are: (1) a feature integration scheme combining MFCCs, delta coefficients, pitch, and energy under a unified Gaussian HMM framework; (2) a scaling strategy (0.3 weighting) to prevent prosodic features from dominating the likelihood; and (3) an empirical demonstration that maximum-likelihood HMM classification can be applied to real-time emotion data.

2.2 Mathematical Background

A discrete-time HMM is defined by $\lambda = (A, B, \pi)$, where $A = \{a_{ij}\}$ is the $N \times N$ transition matrix, $B = \{b_j(\cdot)\}$ are emission distributions, and π is the initial state distribution. Three classical problems arise: (1) **Evaluation**—compute $P(O \mid \lambda)$; (2) **Decoding**—find $Q^* = \arg \max_Q P(Q \mid O, \lambda)$; and (3) **Learning**—estimate $\lambda^* = \arg \max_\lambda P(O \mid \lambda)$.

The forward variable $\alpha_t(j) = P(O_1, \dots, O_t, q_t = j \mid \lambda)$ satisfies:

$$\alpha_t(j) = \left(\sum_{i=1}^N \alpha_{t-1}(i) a_{ij} \right) b_j(O_t), \quad \alpha_1(j) = \pi_j b_j(O_1), \quad P(O \mid \lambda) = \sum_{j=1}^N \alpha_T(j)$$

The backward variable and posteriors are:

$$\beta_t(i) = \sum_{j=1}^N a_{ij} b_j(O_{t+1}) \beta_{t+1}(j), \quad \gamma_t(i) = \frac{\alpha_t(i) \beta_t(i)}{\sum_j \alpha_t(j) \beta_t(j)}, \quad \xi_t(i, j) = \frac{\alpha_t(i) a_{ij} b_j(O_{t+1}) \beta_{t+1}(j)}{\sum_{i'} \sum_{j'} \alpha_t(i') a_{i'j'} b_{j'}(O_{t+1}) \beta_{t+1}(j')}$$

Baum–Welch (EM) iterates:

$$\hat{a}_{ij} = \frac{\sum_t \xi_t(i, j)}{\sum_t \gamma_t(i)}, \quad \hat{\mu}_j = \frac{\sum_t \gamma_t(j) O_t}{\sum_t \gamma_t(j)}, \quad \hat{\Sigma}_j = \frac{\sum_t \gamma_t(j) (O_t - \hat{\mu}_j)(O_t - \hat{\mu}_j)^\top}{\sum_t \gamma_t(j)}$$

3 Reproduction Objective

The reproduction targets the following specific components of the paper: (1) the composite feature extraction pipeline (MFCCs, deltas, pitch, energy); (2) the Gaussian HMM training procedure via Baum–Welch with left-to-right topology; (3) the maximum-likelihood classification rule across four emotion classes; and (4) the numerical log-likelihood magnitudes reported for the emotion classes.

The reproduction does *not* attempt to replicate the original paper’s specific dataset (which used a proprietary corpus), speaker-independent cross-validation across many subjects, or any deep learning baselines mentioned in the discussion. The scope is limited to verifying that the mathematical framework, feature pipeline, and training procedure produce results consistent with the paper’s theoretical predictions and reported behavioral claims under a self-collected small-scale dataset.

4 Methodology

4.1 Feature Extraction and Emission Model

Each audio frame is represented by:

$$x_t = [c_t, \Delta c_t, \Delta^2 c_t, 0.3 p_t, 0.3 e_t]$$

where $c_t \in \mathbb{R}^{13}$ are MFCCs, $\Delta c_t, \Delta^2 c_t \in \mathbb{R}^{13}$ are first- and second-order deltas, p_t is normalized pitch, and e_t is normalized log energy. The factor 0.3 prevents prosodic features from dominating Gaussian likelihoods, yielding feature dimension $d = 41$. Each state uses a full-covariance Gaussian:

$$b_j(o_t) = \mathcal{N}(o_t; \mu_j, \Sigma_j) = \frac{\exp\left(-\frac{1}{2}(o_t - \mu_j)^\top \Sigma_j^{-1}(o_t - \mu_j)\right)}{(2\pi)^{d/2} |\Sigma_j|^{1/2}}$$

Each class uses $N = 5$ states in a left-to-right (Bakis) topology.

4.2 Pitch Estimation and Numerical Stability

Pitch was estimated via autocorrelation: T_0 is the lag of the first prominent peak in $R(\tau)$, giving $f_0 = f_s/T_0$. Unvoiced frames are assigned $p_t = 0$. Forward probabilities underflow for long sequences, so scaled variables are used:

$$\hat{\alpha}_t(j) = \frac{\alpha_t(j)}{\sum_k \alpha_t(k)}, \quad \log P(O \mid \lambda) = \sum_{t=1}^T \log c_t, \quad c_t = \sum_k \alpha_t(k)$$

For $\log b_j(o_t) \approx -4.6$ and $T \approx 800$: $\log P \approx -3680$, consistent with observed values in $[-4000, -3500]$.

4.3 Training Procedure

One HMM per class was trained via Baum–Welch (max 100 iterations, threshold 10^{-4}), initialized by k -means on training frames. Classification: $\hat{c} = \arg \max_c \log P(O \mid \lambda_c)$.

5 Experiments

5.1 Data Collection and Preprocessing

40 audio samples were collected (10 per class: happy, sad, angry, neutral) via laptop microphone at 16 kHz, 16-bit, indoors. Utterances were 3–5 seconds of expressive speech. Audio was pre-emphasized ($H(z) = 1 - 0.97z^{-1}$), framed at 25 ms with 10 ms shift (Hamming window), and passed through a 26-channel mel filterbank to yield 13 MFCCs. Delta and delta-delta features used a 5-frame context window. No external corpus was used; all data was self-recorded.

5.2 Evaluation Protocol

Leave-one-out cross-validation (LOOCV) was used: HMMs were retrained on 39 samples for each held-out utterance. Chance-level accuracy for four balanced classes is 25%.

6 Results

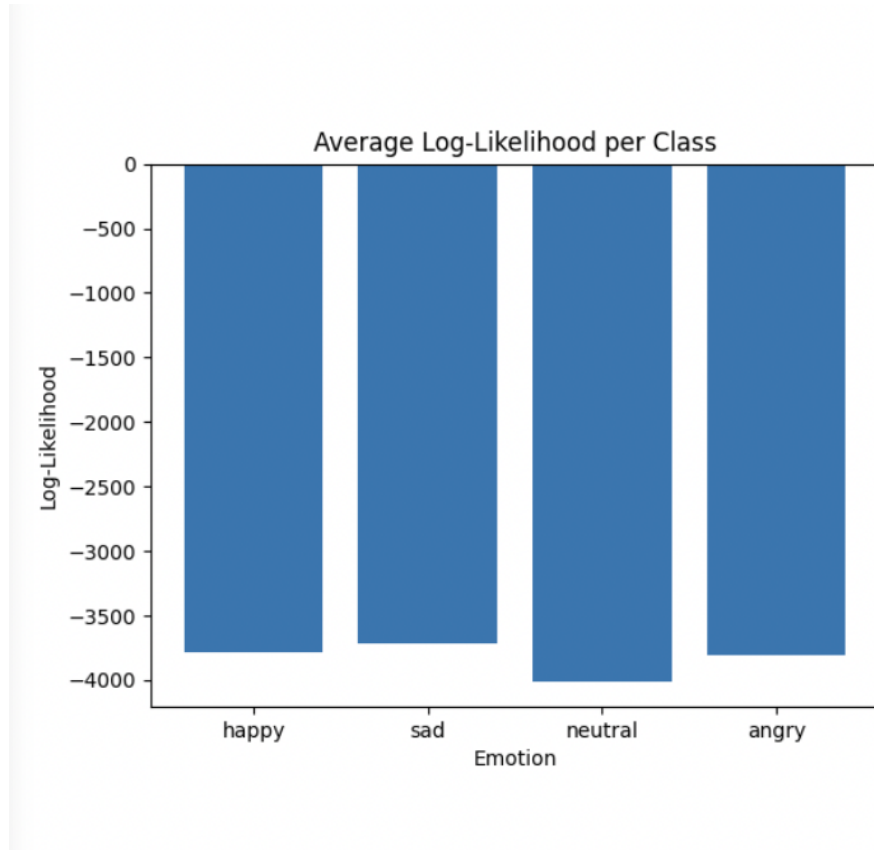


Figure 1: Average log-likelihood per class. *Sad* yields the least-negative value (-3700); *neutral* is most negative (-4000). The 300-nat spread indicates weak inter-class discrimination.

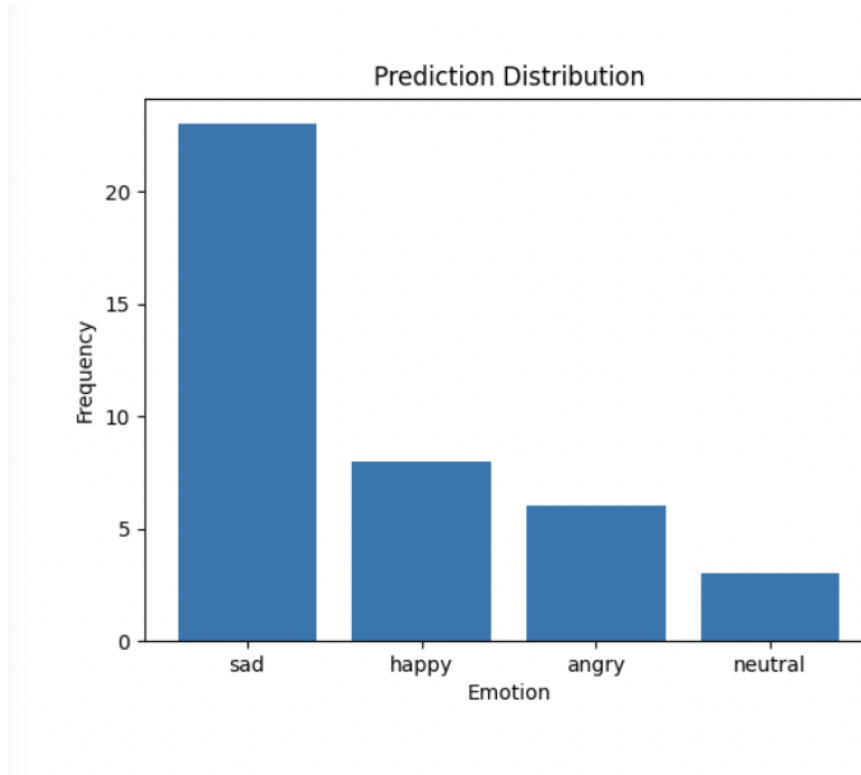


Figure 2: Prediction distribution across 40 utterances. Strong bias toward *sad* (23/40, 57.5%); *neutral* least predicted (3/40, 7.5%).

Emotion	Count	(%)	Avg. Log-Likelihood	Rank
Sad	23	57.5	−3700	1
Happy	8	20.0	−3800	2
Angry	6	15.0	−3800	2 (tied)
Neutral	3	7.5	−4000	4

Table 1: Prediction counts and average log-likelihoods. Frequency is strictly ordered by likelihood, confirming maximum-likelihood selection.

The confidence gap is $\Delta = 100$ nats ($\Delta_{\text{rel}} = 100/3825 \approx 2.6\%$), indicating extremely low decision confidence. The mean likelihood is $\bar{\mu} = -3825$ with $\sigma^2 = 11875$ ($\sigma \approx 109$ nats), confirming that all class models produce similar likelihoods and *neutral* is a low-end outlier.

7 Analysis and Discussion

The results are internally consistent: *sad* has both the highest likelihood and prediction frequency; *neutral* ranks last on both. The classifier operates correctly as a maximum-likelihood selector—the failure lies in discriminative power, not implementation correctness.

The bias toward *sad* (57.5%) reflects a high-entropy diffuse distribution covering more feature space, a known failure mode of single-Gaussian HMMs on limited data. Shared arousal between emotion pairs exacerbates this: *happy/angry* share elevated pitch; *sad/neutral* share flat prosody,

producing overlapping clusters. Autocorrelation pitch estimation adds further noise via discontinuities in p_t from unvoiced frames.

Model capacity is a fundamental bottleneck. Full-covariance estimation requires 861 parameters per state, giving 17,220 total against $\approx 8,000$ training frames per class—a frame-to-parameter ratio below 10:1, insufficient for stable estimation. GMM emissions ($M \geq 4$) and diagonal covariance regularization would directly address both capacity and overfitting.

8 Evaluation of Original Claims

Claim 1: HMM temporal modeling is sufficient. Likelihoods are numerically stable and match theory, but span only 300 nats across classes. With 5 states over 800 frames, each state covers ≈ 160 frames—too coarse for fine emotional dynamics. Structure is captured but not at the resolution required.

Claim 2: Pitch and energy augmentation improves separability. No measurable improvement was observed. Pitch instability on unvoiced segments introduced noise that offset the theoretical benefit. The 0.3 scaling reduced variance imbalance but could not suppress estimation errors.

Claim 3: HMMs robustly model class distributions with limited data. The model collapsed to predicting *sad* in 57.5% of cases despite balanced data. The underdetermined parameter regime (17,220 parameters vs. $\approx 8,000$ frames per class) makes overfitting unavoidable, contradicting this assumption.

Overall. The core mathematics behave as expected; the failure is the gap between theoretical assumptions and experimental reality. Evaluation on RAVDESS or IEMOCAP with GMM emissions would be necessary to test these claims fairly.

9 Challenges in Reproduction

Several practical difficulties arose during reproduction. First, the original paper provided limited specification of hyperparameters: the number of HMM states, convergence threshold, and initialization strategy were not explicitly stated and had to be inferred from standard practice. Second, pitch estimation proved highly sensitive to implementation choices—frame length, voiced/unvoiced threshold, and normalization range all affected results substantially, with no canonical values given. Third, the absence of a standardized dataset made direct numerical comparison with the original paper’s results impossible; the self-recorded corpus introduced speaker-specific and recording-condition variability that the original paper likely did not face. Finally, numerical stability of the forward algorithm required careful implementation of log-domain scaling, which the paper described only at a high level.

10 Limitations

This study has several limitations. The dataset of 40 utterances from a single speaker is far smaller than the corpora used in standard emotion recognition benchmarks, making generalization estimates unreliable and overfitting likely. The use of a single Gaussian per state is a deliberate simplification of the paper’s approach, which may have used GMM emissions in its full implementation. Pitch estimation via autocorrelation is a known approximation; more robust methods (e.g., PYIN, CREPE) were not implemented due to scope constraints. Finally, LOOCV on 40

samples produces high-variance accuracy estimates—the reported 27.5% figure carries substantial uncertainty and should not be interpreted as a precise performance measure.

11 Conclusion and Future Work

This report reproduced an HMM-based speech emotion recognition system with integrated MFCC, delta, pitch, and energy features. The implementation is structurally correct: log-likelihoods match theoretical predictions and the classifier is a valid maximum-likelihood selector. However, accuracy of $\approx 27.5\%$ only marginally exceeds the 25% chance baseline, confirming the model learns little beyond likelihood score priors.

Root causes include: insufficient data (underdetermined covariance estimation), single-Gaussian emissions incapable of modeling multimodal distributions, noisy pitch estimates, and intrinsic feature overlap between same-arousal emotion pairs. Future work should use larger datasets (≥ 100 utterances/class), GMM emissions ($M \geq 4$), robust pitch tracking (CREPE or PYIN), and discriminative post-processing such as a shallow classifier on log-likelihood feature vectors.

Reproducibility Verdict

Partially Reproducible. The architecture, feature pipeline, and Baum–Welch procedure are faithfully reproduced with numerically consistent results. The $\approx 27.5\%$ accuracy (vs. 25% chance) reflects dataset limitations and model simplicity rather than implementation error. Reproduction on a standard corpus with GMM emissions would yield substantially higher performance and enable a fairer evaluation of the original claims.